

Stat4307 Final Exam Question 2 Southpaw

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2023-04-30

```
# Create a data frame with the given information
df <- data.frame(
  district = c(1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4),
  school = rep(1:3, times = 4),
  children = c(130, 150, 160, 110, 120, 100, 89, 130, 100, 100, 120, 150),
  left_handed = c(10, 15, 7, 8, 10, 7, 5, 15, 8, 9, 10, 22)
)

# Calculate the estimated total number of left-handed children in Knoxville
N <- 15 * 3 # total number of schools
n <- 4 # number of clusters sampled
m <- 3 # number of schools per cluster
t1 <- sum(df$left_handed) # sum of left-handed children across all sampled schools
t2 <- sum(df$children) # sum of all children across all sampled schools
t <- (t1 / t2) * N # estimated total number of left-handed children
t # print the estimate
```

```
## [1] 3.886223
```

```
# Calculate the standard error of the estimate
PSU_var <- sum((t1 / t2 - df$left_handed / df$children) ^ 2) / (n - 1)
se <- sqrt(PSU_var / m * (N - n) / (N - 1))

# Calculate the 95% confidence interval
alpha <- 0.05
t_lower <- t - qt(1 - alpha / 2, df = N - n) * se
t_upper <- t + qt(1 - alpha / 2, df = N - n) * se
cat("Estimate of total number of left-handed children in Knoxville:", t, "\n")
```

```
## Estimate of total number of left-handed children in Knoxville: 3.886223
```

```
cat("95% confidence interval: [", t_lower, ",", t_upper, "]\n")
```

```
## 95% confidence interval: [ 3.828078 , 3.944369 ]
```